MIL-R-18401A(AER)
1 JULY 1959
SUPERSEDING
MIL-R-18401(AER)
15 APRIL 1955

MILITARY SPECIFICATION

RADIO SET, AN/ARC-52 AND AN/ARC-52X

This specification has been approved by
the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide two-way radio communication with properly equipped surface stations and other aircraft by means of amplitude modulated radio telephony. When installed with the necessary associated equipment, the Radio Set shall be suitable for the following uses:

(1) Communication by means of A-M radio telephony (A3).

(2) Reception of special type signals consistent with the performance requirements outlined herein.

(3) Automatic direction finding when installed with certain equipment.

(4) Automatic relaying wherein reception of signals on one equipment will cause transmission on another as outlined herein.

1.2 Classification - The equipment covered by this specification shall consist of the following:

Items *	Type Designation	Max. Allow. Size (Inches)			Max. Allow. Wt. (Lbs.)	Appl. Para.
1a- Receiver-Transmitter	RT-332/ARC-52	7-5/8	10-1/4	21-1/2	49.5	3.3.1
1b- Receiver-Transmitter	RT-424/ARC-52X	7-5/8	10-1/4	21-1/2	49.5	3.3.1
2a- Control, Radio Set	C-1607/ARC-52	4-7/8	5-3/4	5-3/4	3.3	3.3.2
2b- Control, Radio Set	C-2072/ARC-52	3-3/8	5-3/4	5-3/4	2.4	3.3.3
3 - Control, Auto Relay	C-2791/ARC	4-7/8	5-3/4	6-3/4	4.3	3.3.4
4 - Mounting	MT-1477/ARC-52	3	11-1/4	21	3.2	3.3.1

* See 6.3

1.3 Auxiliary Equipment - The following equipments are generally similar to those connected to and used with the Radio Set AN/ARC-52 when installed in aircraft. This equipment is not part of the radio set and is not to be furnished by the contractor:

Antenna	AT-141A/A with RG-8/U cable
Headset	H-1/AR or H-4/AR
Microphone	T-17, NAF-213264-6 or ANB-M-C1 carbon type
Throttle Switch	NAF-1124-17

FSC 5821

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Intercommunication Set	AN/AIC-4
Junction Box	J-17A/ARC-5
Jack Box	J-22/ARC-5
Direction Finder Group	AN/ARA-25 or AN/ARA-50

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification.

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-E-7894	Electric Power, Aircraft Characteristics of
MIL-P-17555	Preparation for Delivery of Electronic Equipment, Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-T-18303	Test Procedures, Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect

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STANDARDS

Military

MIL-STD-200	Electron Tubes; and Semiconductor Devices, Diode
MIL-STD-701	Transistors
MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single-Phase AC, or 112/200 Volt 3-Phase AC, or Combinations

2.2 Availability of Documents:

(1) When requesting applicable documents give both the title and number. All requests shall be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, (CDS), Philadelphia 11, Pennsylvania.

2.3 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over all referenced documents.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Deviations from this specification, or from subsidiary documents where applicable, may be made only upon written approval.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts and materials are such their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specification.

(3) When a nonstandard part is used, the meeting of general material requirements such as nonflammable, fungus resistant, nontoxic, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

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(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the Bureau of Aeronautics.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400, except as follows:

(1) Mechanical parts shall be submitted as nonstandard parts only when the material requirements of Specification MIL-E-5400 are not met.

(2) Nonstandard parts of an electrical nature as defined in Specification MIL-E-5400, shall be approved before the preproduction model of the equipment is submitted for tests. The request for approval shall be forwarded to the Bureau of Aeronautics before the part is purchased or fabricated. In addition to or in lieu of the Nonstandard Part Data Sheet, the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Tube Types - The choice and application of electron tubes used in the equipment shall be in accordance with the following requirements and in the sequence noted:

(1) MIL-STD-200 and Navy Supplement -1 thereto

(2) MIL-STD-701

(3) The use of any tube not on these acceptable listings must be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor must submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

3.1.2.1 Electron Tube Complement - A tube complement report shall be prepared in accordance with Specification MIL-E-4682 and submitted to the Bureau of Aeronautics for review and approval prior to preproduction testing. The tube complement shall include all electron tube types (including crystal diodes and transistors) used in the equipment.

3.1.3 Finishes - Brass hardware, shafts, and tuning gears may be used without a protective finish where the part is subject to wear.

3.1.4 Inductors

3.1.4.1 Stability of Q - Every precaution shall be taken to maintain the stability of Q of radio frequency inductors. The requirements shall be only those which enable meeting the equipment performance requirements.

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3.1.4.2 Variable - Mechanical stop provisions in accordance with Specification MIL-E-5400 are not essential where electrical stops are provided and the failure of each stop will not result in damage to the equipment.

3.1.5 Relays - Hermetic seals in accordance with MIL-E-5400 is not required in relays used for latching of positioning mechanisms or for high current applications such as dynamotor starting, provided performance and operation are not compromised, but shall be used in all cases where the state of the art permits.

3.1.6 Switches - Rotary switches mechanically positioned by other than manual means shall be mounted by standoffs in lieu of the requirements of Specification MIL-E-5400.

3.1.7 Soldering (Electrical Connections) - Small lightweight parts and/or tubes employed in printed wiring assemblies processed by dip soldering techniques may use solder for mechanical strength provided adherence to the equipment performance requirements is not compromised.

3.2 Design and Construction:

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient 25°C (+15°C, -5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	27.5 V ±0.5 V DC 115 V ±1 V AC

3.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions for Class 1 equipments or combinations of service conditions as specified in Specification MIL-E-5400, with the following exception:

3.2.3.1 Altitude - The equipment shall operate satisfactorily at an altitude of at least 70,000 feet. Test conditions shall be +10°C ambient temperature, high limit input voltages, and normal duty cycle for a maximum time duration of 30 minutes.

3.2.4 Total Weight - The total weight of the equipment excluding cables shall not exceed 56.0 pounds.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 200 hours, continuously or intermittently, without necessity for readjustment of any controls which are nonaccessible during flight.

3.2.6 Operating Life

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 2000 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

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3.2.7 Interconnecting Cables - This equipment shall be capable of satisfactory operation using external wiring in accordance with Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plug may be easily accomplished.

3.2.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400. However, cables and that portion of connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.9 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the Bureau of Aeronautics prior to preproduction testing.

3.2.10 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in Specification MIL-E-7894 having limits specified herein. The power required shall not exceed the specified amounts.

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(1) RT-332/ARC-52

(a) 115/200 Volt Constant Frequency

Operating Voltage Limits	108 to 121	Volts
Operating Frequency Limits	380 to 420	cps
Reduced Performance Voltage Limits	102 to 108 and 121 to 124	Volts
Power (3ø)	Receive 240	V.A.
	Transmit 390	V.A.
	Channel Change 240	V.A.

(b) DC Power 28 Volts System

Operating Voltage Limits	25 to 29	Volts
Reduced Performance Voltage Limits	22 to 25	Volts
Power Normal	Receive 30	Watts
	Transmit 40	Watts
Current	Receive 1.1	Amps
	Transmit 1.5	Amps

(2) RT-424/ARC-52X

(a) DC Power 28 Volts System

Operating Voltage Limits	25 to 29	Volts
Reduced Performance Voltage Limits	22 to 25	Volts
Power Normal	Receive 288	Watts
	Transmit 414	Watts
Current	Receive 10.5	Amps
	Transmit 15	Amps

3.2.11 Input Power Connections - Input power connections shall be in accordance with MS 25245.

3.2.12 Overload Protection - Overload protection for the equipment shall be provided the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportional to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.12.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.13 Moisture and Fungus Resistant Treatment - Equipments shall be fungus proofed by selection of parts and materials as nonnutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that over-all spraying of the equipment is not necessary.

3.2.14 Identification and Marking - The equipment shall be identified and marked as follows:

3.2.14.1 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of specification MIL-N-18307.

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3.2.14.2 Plug Identification - All plugs shall be suitably identified but not necessarily with the corresponding number designation of the mating part. Plugs which are part of pendant subassembly cables shall be identified only by appropriate marking on the chassis adjacent to the location where they are normally located.

3.2.14.3 Receptacle Identification - The requirement of MIL-E-5400 for the identification of all permanently-mounted receptacles on each side of the panel in a visible location shall apply to the extent that in confined areas only those markings which are practicable are required.

3.2.14.4 Tube Socket Identification - Tube socket identification per MIL-E-5400 is required where space is available. Subassembly covers may be utilized for tube socket and component identification where provision is made to identify the cover with the appropriate subassembly.

3.2.14.5 Reference Symbol Designation - Reference symbol designations shall be in accordance with the requirements of MIL-E-5400, except where space is either limited or not available. Where space is limited, such designations may be made under the component part. Where the foregoing alternatives are impractical, and such designations are omitted, then adequate identification shall be made by means of photographs and/or drawings in applicable instruction handbooks.

3.2.15 Interchangeability - The equipment shall meet the interchangeability requirements as specified in Specification MIL-E-5400. Unless otherwise specified, interchangeability shall be measured against a model supplied by the Bureau of Aeronautics.

3.2.16 Provisions for Maintenance and Field Testing - Provisions for maintenance and field testing shall meet the requirements of specification MIL-E-5400 except that only the antenna input and power connector shall be mounted on the face of the equipment.

3.2.17 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of specification MIL-I-6181.

3.2.18 Terminations

3.2.18.1 Receiver RF - R-F input to the receiver shall be through a 52-ohm cable from a signal generator having an effective source impedance of 52 ohms. Sensitivity shall be in terms of "open-circuit" microvolts.

3.2.18.2 Transmitter RF - The R-F output load (dummy antenna) shall consist of a 52-ohm noninductive load which shall include means suitable for measuring for the R-F power output, percentage modulation and distortion.

3.2.18.3 Receiver Audio - The main (headset) audio output channel of the receiver shall be terminated in a 300-ohm noninductive load which shall include means for measuring the audio-frequency power output. The auxiliary (wide-range) audio output channel of the receiver shall be terminated in a resistance load of 20,000 ohms, including means for measuring the audio output voltage.

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3.2.18.4 Transmitter Input - The audio (microphone) input circuit of the transmitter shall be connected to an audio-frequency generator and dummy microphone having characteristics as listed below:

Carbon Microphone Input

Resistance - 82 ohms
Impedance (mainly resistive at audio frequencies) - 82 ohms
Frequency range - 150-10,000 cps
Total harmonic distortion - less than 1%
Open-circuit output potential (adjustable) 0.05-3.0 V rms

Dynamic Mike Input

Resistance - 82 ohms
Impedance (mainly resistive at audio frequencies) - 82 ohms
Frequency range - 150-10,000 cps
Total harmonic distortion - less than 1%
Open-circuit output potential (adjustable) - 1-50 mv rms

3.2.19 Pressurization - Units of equipment required by other paragraphs of this specification to be sealed shall comply with the appropriate paragraphs of MIL-E-5400 except that a desiccant is not required, a safety valve is not required, mechanical pressurization pumps shall not be built into the equipment and the cases shall be designed to be pumped with dry air to a differential of 2 to 4 pounds per square inch at sea level pressures and room ambient temperatures.

3.2.20 Performance Requirements - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements apply under standard conditions to both main and guard channel operation. Reduced performance and service requirements shall apply only where specifically outlined. Equipment performance under service conditions shall consist of a duty cycle wherein the equipment is operated on a continuous cycle of 10 minutes reception and 5 minutes transmission.

3.2.20.1 Frequency Channels - By means of control box selection, it shall be possible to select any frequency in the range of 225.0 to 399.9 mc in 0.1-mc increments. These main channel frequencies shall correspond with the arithmetic series 225.00, 225.10, 225.20, 225.30-------399.9 mc. Additionally, it shall be possible to select a fixed tuned guard channel in the range of 238.0 mc to 248.0 mc. Unless otherwise specified, the equipment shall be supplied with the guard frequency set for 243.0 mc.

3.2.20.2 Sensitivity - The sensitivity of the receiver shall be such that when an r-f input of 5 microvolts (open-circuit) modulated 30% at 1000 cycles is applied, the audio output power shall be at least 50 milliwatts with a signal plus noise-to-noise ratio (modulated to unmodulated) of 10 db or greater. When the Main Receiver is tuned to frequencies within ±5 megacycles of the guard channel frequency, the main receiver and guard receiver shall meet the sensitivity requirement with 8 microvolts input. Under service conditions the sensitivity shall not be worse than 20 microvolts at any frequency. Under special service conditions of 22 volts dc and 102 volts ac unit power, the signal plus noise-to-noise ratio shall be at least 10 db for 10,000-microvolt signal modulated 30% at 1000 cps.

3.2.20.2.1 Degraded or Unusable Channels Due to Internal Interference - Certain main receiver channel frequencies may be degraded or unusable due to internally generated interference from the guard channel oscillator. When the equipment is supplied with the guard frequency set for 243.0 mc, the main receiver sensitivity measurements shall be made with an input of 20 microvolts at frequencies of 241.1, 241.2 and 275.6 mc. Two unusable channels, 246.6 mc and 253.3 mc are permitted.

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3.2.20.3 AVC Characteristics - The AVC action shall be such that when a 1000-microvolt signal modulated 30% at 1000 cycles is applied, the audio output power shall be 250 milliwatts (±15%), and when the signal input (30% modulated at 1000 cycles) is varied from 10 microvolts to 100,000 microvolts, the audio output power shall not vary by more than ±3 db from the audio output obtained with the 1000-microvolt signal. When a main frequency within 5 mc of the guard frequency is selected, the foregoing requirement need be met only over the range 20 to 100,000 microvolts. Additionally, the receiver shall be protected against blocking under conditions of input signals up to one-half volt. The equipment shall be provided with an internal volume control which shall be adjusted so as to obtain the required output with the 1000-microvolt signal, and then locked before performing other receiver tests.

3.2.20.4 Selectivity - The over-all selectivity of the receivers shall fall within the limits set forth below for either standard or service conditions:

Attenuation Below Resonant Response	Total Bandwidth	
	Guard Receiver	Main Receiver
6 db	60 kc min.	60 kc min.
60 db	400 kc max.	200 kc max.

3.2.20.5 Images and Other Spurious Responses - When receiving on any channel, the receiver shall attenuate images and other spurious responses to the maximum practicable extent and in any case not less than 60 db relative to the desired signal, except that the guard receiver shall be permitted three spurious frequencies attenuated less than 60 db but not less than 40 db.

3.2.20.6 I-F Rejection - Under either standard or service conditions, the receiver shall present an attenuation of not less than 80 db, relative to the response of the desired signal to signals on the frequencies of the intermediate frequency amplifiers.

3.2.20.7 Regeneration - The interstage and over-all regeneration of the receiver shall be so minimized that no evidence of unnecessary oscillation, instability, or excessive regeneration shall occur at maximum r-f gain setting under either standard or service conditions.

3.2.20.8 Noise Peak Limiter - The second detector of the receiver (following the intermediate frequency amplifier) shall be provided with an instantaneous noise peak limiter. The noise limiting action shall effectively limit the audio output voltage of modulating surges or peaks exceeding 50% ±15% modulation of the r-f signal. Its action shall be rapid enough to prevent loss of intelligibility due to reduction in receiver gain under the influence of continuous r-f noises such as are caused by commutators, ignition systems, etc. When a pulsed signal having a pulse duration of 10 microseconds, a pulse repetition frequency of approximately 1000 pulses per second, and peak amplitudes up to 0.1 volt is applied on the same frequency as a desired signal modulated 30% at approximately 1000 cps, not more than 50 microvolts input (desired signal) shall be required to obtain a signal plus noise-to-noise ratio of 3 db.

3.2.20.9 Audio Characteristics - The audio characteristics of the receiver shall be as set forth below. Except where otherwise stated, these requirements apply under both standard and service conditions, and apply to signals detected by either the fixed frequency or variable frequency receivers and amplified in the common headset circuit.

3.2.20.9.1 Capability - Under standard conditions, the receiver shall be capable of delivering an output of at least 250 milliwatts when a signal having any level between 100 and 100,000 microvolts modulated 30% at 1000 cps is applied. With a 1000 microvolt input signal modulated 30% at 1000 cps the total harmonic content of the receiver output shall not exceed 10% at the 250-milliwatt level. Under service conditions the output shall not be more than 6 db below that obtained under standard conditions. Under the special service conditions wherein the d-c supply voltage is 22 volts dc and 102 volts ac, an output of at least 30 milliwatts shall be obtained when a 10,000-microvolt signal modulated 30% [illegible] cps is applied.

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3.2.20.9.2 Over-all Fidelity - The audio response shall not vary more than +1 or -3 db at 300 cps, and +1 or -4 db at 4000 cps relative to the response at 1000 cycles. Modulation frequencies outside the 300-4000 cycles range shall be attenuated the maximum practicable without recourse to inclusion in the receiver of large and/or heavy filter components.

3.2.20.9.3 Auxiliary Circuit - The auxiliary circuits shall deliver only signals detected by the variable frequency receiver. When a 1000-microvolt signal modulated 30% at 1000 cps is applied, the output shall be at least 0.25 volt, and when the modulation frequency is varied from 70 to 7000 cps, the output for any frequency in the range shall be within +2 and -3 db of the output obtained at 1000 cps.

3.2.20.10 Volume Control - Operation of the volume control on the radio set control unit shall cause a minimum of change in the impedance which the radio set presents to external equipment attached to the headset terminals, and such impedance shall not be less than 50 ohms at any setting of the volume control.

3.2.20.11 Signal-to-Noise Type Squelch Characteristics - The main receiver squelch shall be controlled by the audio signal plus noise-to-noise ratio and shall be used with the r-f gain control set for maximum gain. Under conditions of 5 microvolts r-f input modulated at 400 cps, the squelch control provided within the receiver-transmitter shall have a range as to permit squelch operation from no modulation to greater than 30 per cent modulation. With the squelch control adjusted to obtain normal audio output with modulation input required for 6 db signal plug noise-to-noise ratio, the audio shall be effectively suppressed when the modulation input is reduced to a 1-db signal plus noise-to-noise ratio.

3.2.20.11.1 Carrier Type Squelch Circuits - The guard receiver squelch shall be carrier operated and the main receiver shall have alternate soldered squelch circuit linkages for carrier squelch control in equipment automatic relaying applications. For these carrier type squelch circuits, the r-f gain controls provided in the interior of the receiver transmitter shall have such a range as to permit a variation in the sensitivity of the receiver from the maximum to a value of less than 300 microvolts. These controls together with suitable "squelch" circuits provided in the receivers shall permit adjustment so as to effectively suppress unwanted signals or noise in the receiver output under conditions of no, or very low (depending upon the setting of the controls) r-f input signals to the receivers. Under conditions of the r-f gain controls adjusted so as to obtain at least 50 milliwatts output with an r-f input of 6 microvolts 30% modulated at 1000 cycles, the audio output of the receiver shall be effectively suppressed for all r-f inputs, similarly modulated, not in excess of 1.0 microvolt. Selection of a main frequency within 5 mc of the guard frequency shall cause an increase not exceeding two to one in all the foregoing values of microvolts. Operation of the silencer circuit shall be smooth and positive, requiring at its threshold of operation a change of approximately 1 db in r-f input (30% modulated at 1000 cycles) for effecting a change in audio output of 10 db or more.

3.2.20.12 R-F Power Output - The transmitter shall be capable of delivering the following values of radio frequency carrier energy within the range described below:

	225.0 to 399.9 mc
Standard conditions	16 watts min., 20 watts average
Service conditions	9 watts min.
Reduced performance test	500 mw

3.2.20.13 Modulation Characteristics - The transmitter shall meet the following performance requirements for modulation when operated under both standard and service conditions.

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3.2.20.13.1 Radio Telephony (A-3)

3.2.20.13.1.1 Modulation - Under standard conditions the modulator shall be capable of modulating the r-f carrier output of the transmitter not less than 80% when the input voltage at 1000 cps obtained from the dummy microphone (see paragraph 3.2.18.4) is adjusted to not more than 1 volt for the carbon microphone and to not more than 25 millivolts for the dynamic microphone.

Modulation shall be determined from oscilloscope presentation of the modulated carrier wave and from measurement of maximum and minimum voltages displayed on the oscilloscope. Modulation shall be calculated from the formula:

$$\frac{E_{max} - E_{min}}{E_{max} + E_{min}} \times 100\%$$

The modulator of this equipment shall employ clipper or limiter provisions for clipping positive and negative peaks at a modulation level of not more than 95% with increased microphone input voltage.

The modulator gain control shall be adjusted to provide modulation to the clipping level with 1 volt (open circuit) input for the carbon microphone and 25 millivolts (open circuit) input for the dynamic microphone. With the input reduced 3 db below the clipping level, the distortion of 1000 cps modulation envelope shall not exceed 10%. At this input level the transmitter shall have the following fidelity characteristics relative to the output at 1000 cps.

Frequency, cps	Relative Output in db
150	-8 or lower
300	-3 to -7
600	+1 to -2
1,000	0 (ref. level)
3,000	+2 to -2
6,000	+2 to -2
10,000	+1 to -5

3.2.20.13.1.2 Modulation Under Service Conditions - Under service conditions the modulator shall be capable of modulating the r-f carrier output of the transmitter at least 70%. Distortion of the modulation envelope with the input 3 db below the clipping level shall not exceed 10%.

3.2.20.13.1.3 Microphone Current - The microphone energizing current measured with dummy microphone connected and the equipment operated from a 27.5-volt d-c supply shall be 50 ±7 milliamperes dc.

3.2.20.13.2 Tone Modulation - Not less than 70% nor more than 100% modulation of the r-f carrier shall occur when tone transmission is selected and the transmitter is keyed. The frequency of modulation shall be between 920 and 1120 cps.

3.2.20.14 Spurious Radiation - The magnitude of all radiated signals on other than the carrier frequency shall be 40 db or more below the carrier signal, except that harmonic frequencies shall be 30 db or more below the carrier.

3.2.20.15 Frequency Stability - The frequency stability of the complete equipment after operation under either standard or service conditions for five minutes or more shall be such as to maintain the equipment frequencies within the limits shown below provided the temperature does not exceed 55°C:

Transmitter carrier frequency tolerance	±10 kc
Fixed-frequency receiver resonant frequency tolerance	±10 kc
Variable-frequency receiver resonant frequency tolerance	±10 kc

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3.2.20.16 Sidetone - Monitoring of the carrier modulation shall be effected via the receiver audio output channel. The modulation signals delivered to the receiver output while the transmission is in effect shall be obtained by rectifying a small portion of the carrier output rather than from the modulator directly. Under standard conditions the minimum sidetone power output from the receiver shall be approximately 250 milliwatts when the transmitter carrier is modulated 80% at 1000 cycles. Under service conditions the output shall be not less than 75 milliwatts. The total harmonic content of the output shall not exceed 15%.

3.2.20.17 Noise Modulation - Under standard conditions, noise modulation of the transmitter carrier shall be at least 40 db below the maximum modulation obtainable, and the receiver signal plus noise-to-noise ratio shall be at least 30 db when a 10,000-microvolt signal modulated 30% at 1000 cps is applied and the total receiver output (signal plus noise) is adjusted to be 250 milliwatts.

3.2.20.18 Channel Changing Cycle - The time elapsing between the instant the channel selector switch is positioned to select a given channel and the instant the equipment is in an operative condition on the selected channel shall not be greater than six seconds under service conditions. This requirement shall hold for shifting the operating frequency of the equipment between any two available channels. During channel changing operations, the output of the receiver (headset circuit) shall be shorted or otherwise reduced to zero to indicate, through lack of sidetone, that the equipment is not in readiness for transmission should the operator begin speaking before the channel changing cycle is completed.

3.2.20.19 Frequency Modulation - Under the service conditions and with no modulating voltage applied to the transmitter and no r-f signal applied to the receiver, the peak frequency deviation arising from frequency modulation of the receiver conversion oscillator or the transmitter carrier shall not exceed 1.0 kc. Under the standard conditions, and with the emitted carrier modulated 80% at 1000 cycles, the peak frequency deviation arising from frequency modulation of the transmitter shall not exceed 1.0 kc.

3.2.20.20 Transmit-Receive Interval - When shifting from transmission to reception, or vice versa, on the same channel, not more than 300 milliseconds shall elapse between the operation of the control switch and the time the equipment is stabilized in the selected mode of operation in compliance with the performance requirements. This requirement shall be met under service conditions after fifteen minutes operation.

3.2.20.21 Accuracy of Automatic Tuning - When the same frequency channel is selected several times in succession, the transmitter r-f power output and the receiver audio power (with an input signal of 100 microvolts or greater modulated 30% at 1000 cycles) shall not vary by more than .5 db from the power outputs obtained initially. This requirement shall be met under standard or service conditions provided such conditions are static.

3.2.20.22 Cable Length - Under the standard or service conditions that equipment shall operate as required by this specification when using control cables up to 25 feet in length fabricated from #22 AWG copper wire, or when using cables up to 110 feet in length fabricated from #16 AWG copper wire.

3.2.20.23 Relaying - When used in an automatic relaying installation, the receiver-transmitter shall be connected for carbon microphone input and carrier type squelch. When two equipments are connected for automatic relaying, the performance shall be as follows:

3.2.20.23.1 Single-Channel Communication - When transmission and reception only, on either receiver-transmitter, is selected, performance shall be as required by this specification for one receiver-transmitter in a nonrelaying installation.

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3.2.20.23.2 Relaying Operation - The application to the antenna circuit of either radio set of a signal on the receiver resonant frequency and of a level sufficient to fully open the receiver squelch circuit shall cause the other radio set to generate and emit a radio-frequency carrier wave of the rated power on the selected frequency, and removal of the actuating signal shall cause cessation of the generated carrier. When the actuating signal is 3 db or more above the minimum level at which relaying in accordance with the foregoing sentence is possible, the generated carrier shall reach the rated power level within 300 milliseconds after application of the actuating signal and the system shall be returned to standby condition within 300 milliseconds after removal of the actuating signal.

3.2.20.23.3 Relaying Modulation - When the equipment is relaying in accordance with the requirements of the foregoing paragraph and amplitude modulation is applied to the actuating signal, the emitted carrier shall be modulated in an essentially similar manner, but with the peaks clipped by the clipper in the modulator. When a signal 3 db above the level required to actuate the squelch (with sensitivity setting for squelch operation at 5 microvolts) is applied and is modulated 80% at 1000 cycles, the emitted carrier shall be modulated to the extent specified in paragraph 3.2.20.13.1.1.

3.2.20.23.4 Relaying Monitor - When the equipment is relaying as described above, the operator's headset of both equipments shall be connected so as to monitor all signals relayed in either direction. The operator of either equipment can disable the automatic relaying operation and transmit by pressing the push-to-talk switch. Under these conditions, the audio power output may decrease not more than 6 db. Distortion and signal plus noise-to-noise ratio at the terminals of the headset circuit, and r-f carrier output and modulation shall be essentially the same as those obtained when ordinary operation is selected.

3.2.20.23.5 Relay Squelch Change - It shall be possible by changing not more than 7 connections on a suitable terminal board to shift the Receiver Squelch circuits from the signal-to-noise ratio type to the carrier operated type. The carrier operated type is to be used while relaying and the signal-to-noise ratio type is to be used for nonrelay operation.

3.2.20.24 Automatic Direction Finding - When installed with certain associated equipment, the radio set shall be suitable for automatic direction finding. When used for this service, the receiver-transmitter will receive radio frequency signals from the direction finding antenna, amplify and demodulate them, and deliver the low-frequency modulation components to the direction finder amplifying and indicating equipment. When the radio set is connected in an automatic relaying installation, the automatic direction finding function is available on one receiver-transmitter. When the relaying function is selected on the C-2791/ARC, the ADF function is disabled.

3.3 Detailed Requirements

3.3.1 Receiver-Transmitter Unit, RT-332/ARC-52 and RT-424/ARC-52X - The receiver-transmitter shall meet the following requirements.

3.3.1.1 Function - The receiver portion of the equipment shall receive r-f energy from the antenna, amplify and convert it to aural intelligence. The transmitter portion of the equipment shall generate r-f power and transmit it to the load specified herein.

3.3.1.2 Form Factor and Mounting - The receiver-transmitter shall be contained in a single sealed case meeting the applicable requirements of MIL-C-172 and capable of being mounted in an aircraft by means of mounting base MT-1477/ARC-52. It shall be possible to dismount the unit from its mounting base and remove it from the case using ordinary hand tools and without requiring removal of any cables other than the front panel coaxial and power connectors. Mounting base MT-1477/ARC-52 shall be arranged to furnish all support required by the receiver-transmitter unit and it shall provide space between the mounting rails for the entrance air duct

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which is part of the sealed case assembly. When the unit is attached to the mounting and located on a horizontal plane surface, the over-all dimensions shall not exceed the following:

	Height	Width	Length
Over-all dimension of receiver-transmitter and base	8-1/8	11-1/4	21-1/2

3.3.1.3 Contents of the Unit - This unit shall contain the complete transmitter, modulator, receiver, direct crystal-controlled frequency generator, frequency selection system, automatic tuning devices, and all other parts to make the equipment function as a unit except the radio set control specified herein. It shall be possible to remove or replace any plug-in unit located on the main chassis using ordinary hand tools and without unsoldering any connections. It shall be further possible to replace any plug-in unit with another without resynchronization except that realignment of tuning elements may be necessary. The AN/ARC-52 shall contain power conversion facilities which permit operation from the primary power source described in 3.2.10(1)(a) and (b). The AN/ARC-52X shall contain power conversion facilities which permit operation from the primary power source described in 3.2.10.2(a).

3.3.1.4 Weight - The combined weight of the receiver-transmitter and mounting base shall not exceed 52.7 pounds.

3.3.1.5 Electrical Connections

3.3.1.5.1 External - The following electrical connections shall be provided on the front panel of the receiver-transmitter:

(1) Coaxial connector Type N, for connection of receiver-transmitter to the antenna circuits.

(2) A connector conforming to Specification MIL-C-26482 or an approved equivalent shall be provided for connection of power and control circuits.

3.3.1.5.2 Internal - The following electrical connections shall be accessible when the unit is removed from its case:

(1) A Type JK-34, or an approved equivalent, jack for headset connection.

(2) A Type JK-33, or an approved equivalent, jack for microphone connection.

3.3.1.6 Controls, Internal - The following semi-fixed adjustable controls shall be accessible with the unit removed from its case:

(1) Main receiver sensitivity control.

(2) Guard receiver sensitivity control.

(3) Main audio amplifier gain control.

(4) Guard audio amplifier gain control.

(5) Sidetone audio gain control.

(6) Modulator audio gain control.

(7) Signal-to-noise squelch control.

(8) Auxiliary audio gain control.

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3.3.2 Radio Set Control, C-1607/ARC-52 - This control shall be in accordance with Specifications MIL-C-6781 and MIL-P-7788 and shall be suitable for installation in a console panel.

3.3.2.1 Function - This unit shall provide prime control of the AN/ARC-52 Radio Set and shall include all controls for operation and frequency selection of the RT-332/ARC-52 or RT-424/ARC-52X Receiver-Transmitter unit. It shall be possible to manually select any of the available frequencies and preset up to 19 channels on any of these available frequencies. The unit shall have red lighting within the plastic edge lighted plate in accordance with specification MIL-P-7788.

3.3.2.2 Form Factor and Mounting - The over-all dimensions of this unit shall not exceed the following:

	Height	Width	Depth
Over-all dimensions of control box	4-7/8	5-3/4	5-3/4

3.3.2.3 Weight - The weight of this unit including the mating connector shall not exceed 3.3 pounds.

3 3.2.4 Contents of the Unit - This control unit shall contain all circuits necessary to control manual frequency or preset channel selection, emission, ADF operation and primary power to the equipment.

3.3.2.5 Electrical Connections - All electrical connections to this unit shall be made through a connector conforming to Specification MIL-C-26482 or an approved equivalent.

3.3.2 6 Controls, Panel Mounted - The following controls shall be located and suitably marked on the front panel.

(1) A 20 position rotary switch-type control "Chan" for selecting any one of 18 predetermined main frequencies, or the guard frequency, or a manual channel on which any of the 1750 available frequencies can be selected.

(2) A 4-position rotary switch for controlling all primary input power to the equipment and for selecting the mode of transmission and reception as follows; Position One-"OFF;" Position Two-"T/R" operation on frequency selected by (1) above. Position Three-"T/R+G"-guard receiver operation in addition to that of Position Two; Position Four-"ADF" operation on frequency selected by (1) above.

(3) A rotary volume control "VOL" for adjusting the amplitude of the audio signals delivered to the headset circuit.

(4) Four controls "MANUAL" calibrated in frequency for manually selecting any of the 1750 frequencies.

(5) Means for readily setting up new predetermined frequencies on the 20-position switch type control.

3.3.2.7 Vibration - Maximum vibration acceleration to which this control unit shall be subjected to is ±8G.

3 3.3 Radio Set Control, C-2072/ARC-52 - This control shall be in accordance with Specifications MIL-C-6781 and MIL-P-7788 and shall be suitable for installation in a console panel.

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3.3.3.1 Function - This unit shall provide prime control of the AN/ARC-52 Radio Set and shall include all controls for operation and frequency selection of the RT-332/ARC-52 or RT-424/ARC-52X Receiver-Transmitter unit. It shall be possible to preset up to 20 channels on any of the available frequencies. The unit shall have red lighting within the plastic edge lighted plate in accordance with specification MIL-P-7788.

3.3.3.2 Form Factor and Mounting - The over-all dimensions of this unit shall not exceed the following:

	Length	Width	Depth
Over-all dimensions of control box	3-3/8	5-3/4	5-3/4

3.3.3.3 Weight - The weight of this unit including the mating connector shall not exceed 2.4 pounds.

3.3.3.4 Contents of the Unit - This control unit shall contain all circuits necessary to control preset channel selection, emission, ADF operation and primary power to the equipment.

3.3.3.5 Electrical Connections - All electrical connections to this unit shall be made through a connector conforming to Specification MIL-C-26482 or an approved equivalent.

3.3.3.6 Controls, Panel Mounted - The following controls shall be located and suitably marked on the front panel.

(1) A 20-position rotary switch-type control "Chan" for selecting any one of 19 predetermined main frequencies, or the guard frequency.

(2) A 4-position rotary switch for controlling all primary input power to the equipment and for selecting the mode of transmission and reception as follows; Position One- "OFF;" Position Two- "T/R" operation on frequency selected by (1) above; Position Three- "T/R+G" -guard receiver operation in addition to that of Position Two; Position Four- "ADF" operation on frequency selected by (1) above.

(3) A rotary volume control "VOL" for adjusting the amplitude of the audio signals delivered to the headset circuit.

(4) Means for readily setting up new predetermined frequencies on the 20-position switch type control.

3.3.3.7 Vibration - Maximum vibration acceleration to which this control unit shall be subjected to is ±8G.

3.3.4 Automatic Relay Control C-2791/ARC - This control shall be in accordance with Specifications MIL-C-6781 and MIL-P-7788 and shall be suitable for installation in a console panel.

3.3.4.1 Function - This unit shall provide prime control of the AN/ARC-52 Radio Set and shall include all controls for operation and frequency selection of the RT-332/ARC-52 or RT-424/ARC-52X Receiver-Transmitter unit when used with a C-1607/ARC-52 in an automatic relaying installation. During automatic relaying operation the headset audio is common for #1 and #2 equipments. Volume controls on the control units adjust the individual headset audio levels. The operator of either equipment can disable the automatic relaying operation and transmit by pressing the push-to-talk switch. It shall be possible to manually select any of the available frequencies and preset up to 19 channels on any of these available frequencies. The unit shall have red lighting within the plastic edge lighted plate in accordance with specification MIL-P-7788.

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3.3.4.2 Form Factor and Mounting - The over-all dimensions of the unit shall not exceed the following:

	Height	Width	Depth
Over-all dimension of control box	4-7/8	5-3/4	6-3/4

3.3.4.3 Weight - The weight of this unit including the mating connector shall not exceed 4.3 pounds.

3.3.4.4 Contents of the Unit - This control unit shall contain all circuits necessary to control manual frequency or preset channel selection, emission, ADF operation, automatic relaying operation and primary power to the equipment.

3.3.4.5 Electrical Connections - All electrical connections to this unit shall be made through a connector conforming to Specification MIL-C-26432 or an approved equivalent.

3.3.4.6 Controls, Panel Mounted - The following controls shall be located and suitably marked on the front panel.

(1) A 20 position rotary switch-type control "CHAN" for selecting any one of 18 predetermined main frequencies, or the guard frequency, or a manual channel on which any of the 1750 available frequencies can be selected.

(2) A 5-position rotary switch for controlling all primary input power to the equipment and for selecting the mode of transmission and reception as follows; Position One- "OFF"; Position Two- "T/R" operation on frequency selected by (1) above; Position Three- "T/R+G" guard receiver operation in addition to that of Position Two; Position Four-"ADF" operation on frequency selected by (1) above; Position Five- "REL" operation on frequency selected by (1) above.

(3) A rotary volume control "VOL" for adjusting the amplitude of the audio signals delivered to the headset circuit.

(4) Four controls "MANUAL" calibrated in frequency for manually selecting any of the 1750 frequencies.

(5) Means for readily setting up new predetermined frequencies on the 20-position switch type control.

3.3.4.7 Vibration - Maximum vibration acceleration to which this control unit shall be subjected to is ±5 G.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

(3) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4.2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-T-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181. The sampling and testing of plastic lighting plates shall be in accordance with Specification MIL-P-7788.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 30 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

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(2) Vibration Test (90 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life tests shall be performed under the conditions specified in 4.4.1. The life test sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1
201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.4.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room temperature
Altitude	Normal room (0 to 5,000 feet)
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	26.5 ± 2.5 volts

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

4.4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

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(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.4.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life tests shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the Bureau of Aeronautics.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Bureau of Aeronautics.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. ECP's, MC's, and MB's shall be prepared in accordance with SAR-300.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request shall be accompanied by complete supporting information.

6.3 Ordering Data - Procurement documents should specify the following:

(1) List of items of 1.2.

(2) Selection of applicable levels of preservation, packaging and packing.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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